

Automated data-integrity checks

SCR data checker app

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Session objectives

Observation: unstandardized, undocumented data entry practices introduce unnecessary errors and take a lot of time to correct.

Main goal: check common data problems before building traps and capthist objects for secr.

1. Show where the app fits in the workflow
2. Summarize the main checks on traps and detections csv files
3. Show how to launch the app and use the report
4. Leave the details to the live demo

Where this app fits

Key idea: this is an initial check **before** the main SECR scripts.

- Upload a traps csv and a detections csv
- Let the app propose standard column names
- Approve the mapping if it looks right, else manually revise
- Review errors/warnings, correct where needed
- Download an HTML report for record-keeping

What the app expects

The app is built around the standard workshop inputs:

- `traps.csv`: detector locations, effort, and optionally dates/covariates
- `detections.csv`: animal detections by session, occasion, and trap

It first checks column names and can map common alternatives such as:

- `trap id`, `camera id`, `station` → `trapID`
- `easting`, `northing` → `x`, `y`
- `latitude`, `longitude` → `lat`, `long`
- `animal id`, `individual id` → `animalID`

If no `session` column is found, the app can create `session = 1`.

Checks on the traps file

Tasks involved:

- validate identifiers
- validate coordinates
- validate effort
- compare dates against effort
- summarize extra covariates

Checks on the traps file

Main checks:

- `session` must be a positive integer, or text with no spaces
- `trapID` must be unique within session
- `trapID` with spaces or punctuation is flagged as a warning
- coordinates must be numeric, with no comma decimals or stray whitespace
- at least one coordinate pair must exist: `x/y` or `lat/long`
- duplicated coordinates used by different trap IDs are flagged
- possible swapped coordinates are flagged
- `effort` must be numeric and strictly positive
- `start_date` and `end_date` are parsed and compared with `effort`

Checks on the detections file

Tasks involved:

- validate IDs and occasions
- parse dates
- compare detections against traps
- review optional covariates

Checks on the detections file

Main checks:

- `session` must be a positive integer, or text with no spaces
- `animalID` must not contain spaces or punctuation
- placeholder IDs such as `unknown`, `unk`, `unsure`, `na` are flagged
- alternate forms of the same ID, such as notes in brackets, are flagged
- `occasion` must be a positive integer
- `date` is parsed and impossible dates are flagged
- overlapping date ranges across occasions are flagged
- optional `sex` values outside `Male/Female` or `M/F` are flagged

Cross-file checks

Main checks:

- detections whose `trapID/session` combination is missing from `traps.csv`
- detections at traps with `effort <= 0`
- detection sessions missing from `traps.csv`
- detection dates outside the recorded trap operating period
- sessions present in `traps.csv` but absent from detections, shown in the summaries

Also gives quick summaries, plots, and maps to help spot problems visually.

Installation and launch

Use the launcher script (recommended):

```
source("data_checker/run_data_checker.R")
```

Or from the course_materials R project:

```
install.packages(c(
  "shiny", "shinydashboard", "DT", "ggplot2", "dplyr", "readr",
  "stringr", "tidyr", "scales", "leaflet", "sf", "rmarkdown", "htmltools"
))

shiny::runApp("data_checker")
```

How to use it

Brief workflow for the demo:

1. Upload `traps.csv` and `detections.csv`, or pick one of the example files
2. Check the proposed column-name mapping and approve it
3. (Optional) Enter CRS `+proj=utm +zone=44 +datum=WGS84 +units=m +no_defs` for map display
4. Open the Data Checks tab and work through errors before warnings
5. Download the HTML report once the inputs look sensible

Key idea: fix input problems here, before they turn into harder-to-interpret errors later in `seccr`.

Live demo

- Upload a traps csv and a detections csv with required column names
- If no exact match, app will propose which columns to use (but no renaming, yet)

SOR Data Checker

Column Names

Data Checks

Uploads and Column Approval

Upload traps csv

Browse... No file selected

Upload detections csv

Browse... No file selected

Optional CRS

EPSG:32644

Or choose example traps csv

None

Or choose example detections csv

None

Traps column mapping

original	proposed	exact
Camera ID	trapID	false
Latitude	lat	false
Longitude	long	false
UTM Easting	x	false
UTM Northing	y	false
UTM Zone		
Installation Date	start_date	false
Last Operational Date	end_date	false
Effort (days)	effort	false
Topography		
Altitude		
Water present at location		
Note		
<<create session>>	session	false

Approve Reject

Detections column mapping

original	proposed	exact
Session	session	true
Snow Leopard ID	animalID	false
Occasion	occasion	true
Date of detection (MDY)	date	false
Camera ID	trapID	false
Sex		

Approve Reject

- Data Checks tab shows the main warnings and errors from both files.

The screenshot displays the 'SCR Data Checker' application interface. On the left is a dark sidebar with a menu containing 'Column Names' and 'Data Checks'. The 'Data Checks' option is selected. The main content area has a blue header with the title 'Report'. Below the header, a green message states: 'Both files are approved. Detailed checks and report download are available below.' A button labeled 'Download HTML report' is positioned below this message. The interface is divided into two panels: 'Traps File' and 'Detections File'. Each panel shows 'Errors: 0 | Warnings: 1'. The 'Traps File' panel has a table with columns: session, trapID, Coordinates, Effort, Dates, and Covariates. A warning is listed: 'Warning trapID contains whitespace or punctuation that may not parse reliably in secr. Problem values: Camera 8, Camera 11, Camera 1A, Camera 12, TPC 48, TPC 18, TPC 23, TPC 26, ...'. Below the table, it says 'Data type: numeric'. The 'Detections File' panel has a table with columns: session, animalID, occasion, trapID, Date, and Covariates. Two warnings are listed: 'Error Some detections reference a trapID/session combination not found in traps.csv. Problem values: 1:TPC37, 1:TPC1A, 1:TPC31, 1:TPC25, 1:TPC32, 1:ANICID296, 1:Camera10, 1:TPC11, ...' and 'Warning Sex contains values other than Male/Female or N/A. Problem values: Unknown'. Below the table, it says 'Data type: numeric'.

- Bad example files are provided to demonstrate the kinds of problems the app catches.

The screenshot displays the 'SCR Data Checker' application. On the left is a dark sidebar with 'Column Names' and 'Data Checks' options. The main area is titled 'Report' and contains a message: 'Both files are approved. Detailed checks and report download are available below.' with a 'Download HTML report' button. Below this are two panels: 'Traps File' and 'Detections File'. Each panel shows a table header and a list of errors/warnings.

Traps File

Errors: 2 | Warnings: 2

session	trapID	Coordinates	Effort	Dates	Covariates
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- Warning trapID contains whitespace or punctuation that may not parse reliably in secr. Problem values: Camera&11, TPC 38
- Warning Possible swapped coordinates detected for: Camera&11
- Error start_date could not be parsed for all rows. Some values do not match the inferred date format. 1312/2022, 2023/28/05
- Error end_date could not be parsed for all rows. Some values do not match the inferred date format. 2023/29/06

Data type: numeric

Detections File

Errors: 4 | Warnings: 2

session	animalID	occasion	trapID	Date	Covariates
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- Error animalID must not contain whitespace or punctuation. Problem values: SL_12, SL_14
- Error Some animal IDs appear with trailing notes or alternate punctuated forms. Problem values: SL12, SL_12, SL_14, SL14
- Error Some detections reference a trapID/session combination not found in traps.csv. Problem values: 1:TPC11, 1:TPC35555, 11:TPC11
- Error Detections contain sessions not present in traps.csv. Problem values: 11
- Warning Some detection dates fall outside the operational period recorded for the trap. Problem values: 1:TPC14:2021-09-12
- Warning Sex contains values other than Male/Female or M/F. Problem values: Unknown

Data type: numeric